STAFF REVIEW · H446 239 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Cover colour does not determine availability.

2. A function/procedure with a clear interface usable in multiple places.

3. 49.

4. Squares of the same integer do not change; attendance does.

I Do — source worked steps

1. Predict first — The key A is already stored, so the lookup uses the cache.

2. Model and check — If pricing changes, the cached price must be updated or expired.

We Do — guided CHECK

1. Predict first
Using the old cached 12 gives a stale answer instead of the new total 15.
Model and check
Update or invalidate the cached value when attendance changes, or apply a suitable expiry policy.

You Do — source mark schemes

1. [4] Input 6 [1]; output 36 [1]; valid integer input [1], no overflow for chosen integer type [1].

2. [6] First misses [1], computes 49 [1], stores 7→49 [1]; second hits [1], returns 49 [1] without recomputing [1].

3. [6] Faster repeated lookups [1], reduced repeated database work [1], uses memory [1], can return stale information [1], stale data can cause incorrect decisions [1], update/invalidate or expire on change [1].

Exit Ticket — source answer

1. False. Dynamic data can change, leaving a stale result.